Python Programmation

CM 7 : Second Course Recap

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During this recap, we will correct exercises
from previous practical sessions.

Short exercises: using the board.
Long exercises: by group (10 groups max).

CM5 Doggy bag

File handling

- Open and write a file
- Encoding and pathing
- String parsing
- Dedicated libraries

File navigation

- File VS python structures
- Common errors



CM6 Doggy bag

Libraries

- The toolboxes of python
- pandas, sklearn and matplotlib

Virtual env

- Use them !
- Global venv = BAD

Modules

- Snippets of code
- Better code structure



Let’s proceed to the corrections !

Now, let's play a game…
Can you guess the result of the following scripts ?

First Guess

```
data = [
    "file1",
    "file2"
    "file3"
]

print (len (data))
```

First Guess

```
data = [
    "file1",
    "file2"
    "file3"
]

print (len (data))
```

- No error, but this list only has two elements
- Do not forget the commas !

Second Guess

```
def f (x):
    return (x+y)

y = 2
print (f(3))
```

Second Guess

```
def f (x):
    return (x+y)

y = 2
print (f(3))
```

- No error, return 5
- Refer to python scopes if needed

Third Guess

```
if 2 < 4 > 3 < 5:
    print("simple")

if 2 < 4 > 3 < 5:
    print("simple")
```

Third Guess

```
if 2 < 4 > 3 < 5:
    print("simple")

if 2 < 4 > 3 < 5:
    print("simple")
```

- Will return "simple" and nothing, respectfully
- To read from left to right

Fourth Guess

```
a = [print(i) for i in range(10)]
print(a)
```

Fourth Guess

```
a = [print(i) for i in range(10)]
print(a)
```

- Will print(i) each time, then a list of None
- use `"".join()` to print a list instead

Fifth Guess

```
a = 100
b = 10**2
print(a is b)

a = 1000
b = 10**3
print(a is b)
```

Fifth Guess

```
a = 100
b = 10**2
print(a is b)

a = 1000
b = 10**3
print(a is b)
```

- Print True, then False
- Why ? Take your guesses

Sixth Guess

```
a = 0
b = 0

while a < 1000:
    check = a is b
    print(check)
    a += 1
    b += 1
```

Sixth Guess

```
a = 0
b = 0

while a < 1000:
    check = a is b
    print(check)
    a += 1
    b += 1
```

- Print True up to 256
- Numbers from -5 to 256 have dedicated memory space

Seventh Guess

```
x = 'abc'
y = 'abc'
print(x is y)

x = 'abc!'
y = 'abc!'
print(x is y)
```

Seventh Guess

```
x = 'abc'
y = 'abc'
print(x is y)

x = 'abc!'
y = 'abc!'
print(x is y)
```

- Print True, then False
- Some strings have dedicated memory space

Eight Guess

```
with open("toto.txt",'r') as f:
    for x in f:
        print(x)
```

Eight Guess

```
with open("toto.txt",'r') as f:
    for x in f:
        print(x)
```

- Print each line of "toto.txt"
- The iretator of an object is not always clear